TEACHER GUIDE DRAFT

IB Diploma Programme Business management HL

# Production and capacity

Unit 5: Operations management
**5.2 Operations methods; 5.6 Production planning (HL only)**



A revision lesson. Pupils run a car dealership in firms of four and revise production and capacity on their own deals. Seat each firm together. Each firm answers the topic tasks together.

## The lesson

**45-minute setting**

| Setup | Round 1 | Round 2 | Round 3 | Final questions |
|---|---|---|---|---|

### EVERY ROUND

**Firm decisions**
3 min

→

**Deals**
4½–5 min

→

**Topic tasks**
4 min

→

**B2B deal**
2½ min
from Round 2

### LESSON SETTINGS

30 min | **45 min** | 60 min | 75 min

# What pupils do

Every pupil buys and sells cars. In Firm decisions, each role has its own job.



## CEO

Staff · Sabotage from Round 2



## Finance

Capital & Loans · Investments



## Sales & Marketing

Marketing



## Operations

Seasonal stock

# Topic tasks

Each firm gets its own figures. The method is the same. Your Live screen shows each task and every firm's answer.

## 1 Capacity used %

Round 1

Workshop capacity.

**Capacity used, as a %?**

| | |
|---|---|
| Prep bays | **3 bays** |
| Most one bay can do a week | **10 cars** |
| Cars prepared a week | **24 cars** |

**ANSWER**

**80%**

3 × 10 = 30; 24 ÷ 30 × 100 = 80%

**Why:** It has 6 cars of spare capacity a week.

## 2 Full capacity risks

Round 1

You run at 100% capacity all year.

**Which two problems are most likely?**

- **A** Spare capacity is wasted
- **B** Staff stress and more errors
- **C** A higher fixed cost per car
- **D** No spare time for repairs
- **E** Staff with too little to do

**ANSWER**

**✓ Staff stress and more errors**
**✓ No spare time for repairs**

**Why:** Full capacity spreads fixed costs well, but leaves no slack.

## 3 Fixed cost per car

Round 2

Three workshops: fixed costs a year, and cars prepared.

**Order by fixed cost per car, lowest first?**

- **A** £45,000 fixed, 500 cars
- **B** £20,000 fixed, 160 cars
- **C** £30,000 fixed, 300 cars

**ANSWER**

**1. £45,000 fixed, 500 cars**
**2. £30,000 fixed, 300 cars**
**3. £20,000 fixed, 160 cars**

**Why:** £90, £100, £125 a car: the biggest costs, spread widest, are cheapest.

# Topic tasks

## 4 New productivity

Round 2

More staff, more cars.

**New cars prepared per member of staff?**

| | |
|---|---|
| Cars prepared a week | **18 cars** |
| Staff | **3 staff** |
| Extra staff | **3 staff** |
| Extra cars prepared a week | **30 cars** |

**ANSWER**

**8 cars**

(18 + 30) ÷ (3 + 3) = 8 cars

**Why:** Productivity rose from 6 to 8 cars per worker.

## 5 Meet a short peak

Round 3

Demand is above your capacity for two months only.

**How should you meet it, and why?**

**A** Build a new workshop

**B** Subcontract the extra work

**ANSWER**

**✔ Subcontract the extra work**

**Why:** A short peak does not pay for new capacity.

## 6 Unit cost when full

Round 3

Run the workshop full.

**Fall in unit cost at full capacity?**

| | |
|---|---|
| Fixed costs, a month | **£12,810** |
| Variable cost per car | **£400** |
| Cars prepared a month | **15 cars** |
| Most it can prepare a month | **35 cars** |

**ANSWER**

**£488**

£12,810 ÷ 15 = £854; ÷ 35 = £366; £854 − £366 = £488

**Why:** The variable cost stays; only the fixed cost per car falls.

# Final questions

Each pupil answers 6 final questions on production and capacity. They come from these 10 questions.

Then each pupil writes one short answer: “Evaluate hiring staff or paying overtime to raise output.”

1 A service centre can complete 640 jobs a month but completes 512. What is its capacity utilisation?
✔ **80%**

2 A parts team produces 1,440 tested units in 240 labour hours. What is labour productivity?
✔ **6 units per labour hour**

3 A dealer considers outsourcing parts deliveries. Which comparison best supports an operational decision?
✔ **Total cost, reliability, lead-time risk and capability kept or lost**

4 A bodyshop's practical monthly capacity is 900 repair-hours and booked work uses 765 hours. What is spare capacity as a percentage of practical capacity?
✔ **15%**

5 A workshop raises utilisation from 82% to 98%, while waiting time and rework rise sharply. What is the best interpretation?
✔ **It is near its limit, so delays rise**

6 Technician productivity differs widely between branches. Which study is most likely to support a fair operational decision?
✔ **Compare job mix and hours first**

7 Before a wellbeing redesign a workshop produced 450 standard hours with 90 paid hours; after, 520 with 104. What happened to measured labour productivity?
✔ **It remained 5 standard hours per paid hour**

8 A fitting centre’s 27 fitters fit 918 tow bars a month. What is labour productivity?
✔ **34 tow bars per fitter**

9 A bodyshop can do 250 repairs a month and did 160. What is its capacity utilisation, and the effect on unit costs?
✔ **64%: each repair carries more fixed cost than at full capacity**

10 A workshop can service 400 vehicles a month and serviced 300 last month: 75%. Which measure is that?
✔ **Its capacity utilisation**